

The matching/permanent bridge:
 $\pi(T) = \text{per } L(T) / \prod \deg$ as a weighted matching sum, and its
cavity recursion

Abstract

This note supplies the graph-theoretic bridge underlying the Laplacian-ratio program: the fact that the amplitude quantity $\pi(T) = \text{per } L(T) / \prod_v \deg(v)$ used throughout equals the value computed by the cavity recursion on the rooted tree. It has two halves, both proved here in full: **(H1)** for any tree (indeed any forest) $\text{per } L(T) = \sum_M \prod_{v \notin V(M)} \deg(v)$, the sum over matchings M of the product of degrees of the unmatched vertices — a consequence of acyclicity (permutations contributing to the permanent are exactly the involutions whose transpositions are edges); and **(H2)** dividing by $\prod \deg$ gives $\pi(T) = \sum_M \prod_{ij \in M} \frac{1}{\deg(i) \deg(j)}$, the edge-weighted matching partition function, which satisfies the standard tree cavity recursion $r_v = (1 + \sum_c w_{vc} r_c)^{-1}$ with $w_{ij} = \frac{1}{\deg(i) \deg(j)}$. Both identities are verified numerically to machine precision on trees up to 10 vertices, and the algebraic recursion step (H2) is machine-checked in Lean 4 (`formalization/R3Cert/Matching.lean`). Together they establish, for the *raw* Laplacian weights $w_{ij} = 1/(\deg i \deg j)$, that the edge-weighted cavity recursion telescopes to exactly $\pi(T)$. The separate identification of the DEC-*dressed* R3 branch amplitude $\log \Phi$ with $\pi(T)$ is a distinct bridge (companion Steps 1–4): Steps 1–2 (cherry-folding) are machine-checked, Steps 3–4 (the SimpleGraph realisation and the ρ_B/hub -limit amplitude normalisation) remain open, so it is *not* settled here.

1 Setup

Let T be a tree on vertex set V , $|V| = n$, with Laplacian $L = L(T)$: $L_{vv} = \deg(v)$, $L_{uv} = -1$ if $uv \in E(T)$, and $L_{uv} = 0$ otherwise. The Laplacian ratio is

$$\pi(T) = \frac{\text{per } L(T)}{\prod_{v \in V} \deg(v)}, \quad \text{per } A = \sum_{\sigma \in S_V} \prod_v A_{\sigma(v), v}.$$

A *matching* $M \subseteq E(T)$ is a set of pairwise disjoint edges; $V(M) = \bigcup_{e \in M} e$ is the set of matched vertices. The empty matching is allowed.

2 (H1) The permanent is a matching sum

Theorem 1. *For every forest T ,*

$$\text{per } L(T) = \sum_{M \text{ matching}} \prod_{v \notin V(M)} \deg(v).$$

Proof. Expand $\text{per } L = \sum_{\sigma \in S_V} \prod_v L_{\sigma(v), v}$. Fix σ and suppose its term is nonzero; then $L_{\sigma(v), v} \neq 0$ for every v , i.e. for each v either $\sigma(v) = v$ or $\sigma(v)v \in E(T)$.

Claim: σ is an involution. Write σ as a product of disjoint cycles and take any cycle $(v_0 v_1 \dots v_{k-1})$ with $\sigma(v_i) = v_{i+1}$ (indices mod k). If $k \geq 3$, the v_i are distinct and each $v_i v_{i+1} \in E(T)$ (a fixed point would be a 1-cycle; a nonzero factor with $\sigma(v_i) \neq v_i$ forces the edge), so $v_0 v_1 \dots v_{k-1} v_0$ is a cycle in T — impossible, as T is acyclic. Hence every cycle has length 1 or 2, i.e. $\sigma^2 = \text{id}$.

Thus the 2-cycles of σ are transpositions $(u w)$ with $uw \in E(T)$, and being disjoint they form a matching M_σ ; the fixed points are exactly $V \setminus V(M_\sigma)$. Conversely every matching M determines the involution σ_M swapping the endpoints of each edge of M and fixing the rest, and $M_{\sigma_M} = M$; so $\sigma \mapsto M_\sigma$ is a bijection between the nonzero-term permutations and the matchings. Finally, for $\sigma = \sigma_M$,

$$\prod_v L_{\sigma(v),v} = \prod_{v \notin V(M)} L_{vv} \cdot \prod_{(u,w) \in M} L_{wu} L_{uw} = \prod_{v \notin V(M)} \deg(v) \cdot \prod_{uw \in M} (-1)(-1) = \prod_{v \notin V(M)} \deg(v),$$

since each transposition contributes $L_{uw} L_{wu} = 1$. Summing over σ (equivalently over M) gives the claim. $\square$

Corollary 2. $\pi(T) = \frac{\text{per } L(T)}{\prod_v \deg(v)} = \sum_M \prod_{v \in V(M)} \frac{1}{\deg(v)} = \sum_M \prod_{uw \in M} \frac{1}{\deg(u) \deg(w)}.$

Proof. Divide Theorem 1 by $\prod_v \deg(v)$: $\prod_{v \notin V(M)} \deg(v) / \prod_v \deg(v) = 1 / \prod_{v \in V(M)} \deg(v)$, and $V(M)$ is the disjoint union of the edge endpoints of M . $\square$

So $\pi(T)$ is the *edge-weighted matching partition function* $Z(T) = \sum_M \prod_{e \in M} w_e$ with $w_{uw} = \frac{1}{\deg(u) \deg(w)}$.

3 (H2) The matching sum satisfies the cavity recursion

Root T at a vertex ρ . For a vertex v let T_v be its subtree, and define the two partition functions

$$P_v = \sum_{M \text{ matching of } T_v} \prod_{e \in M} w_e, \quad P_v^0 = \sum_{\substack{M \text{ matching of } T_v \\ v \notin V(M)}} \prod_{e \in M} w_e,$$

the full sum and the sum restricted to matchings leaving v unmatched. Set the cavity ratio $r_v = P_v^0 / P_v$.

Lemma 3. *For a vertex v with children $c_1, \dots, c_k$ in the rooted tree,*

$$P_v^0 = \prod_{i=1}^k P_{c_i}, \quad P_v = \left(\prod_{i=1}^k P_{c_i} \right) \left(1 + \sum_{i=1}^k w_{vc_i} r_{c_i} \right), \quad r_v = \frac{1}{1 + \sum_{i=1}^k w_{vc_i} r_{c_i}}.$$

(For a leaf, $k = 0$: $P_v^0 = P_v = 1$, $r_v = 1$.)

Proof. A matching of T_v leaving v unmatched is exactly a choice of a matching in each child subtree T_{c_i} , independently; hence $P_v^0 = \prod_i P_{c_i}$ (weights multiply over the disjoint subtrees). A matching using v must match v to exactly one child c_i (edges at v go only to its children), contributing weight w_{vc_i} , forcing c_i unmatched inside T_{c_i} (weight $P_{c_i}^0$) while the other child subtrees are matched freely (weight $\prod_{j \neq i} P_{c_j}$). Summing over the choice of c_i and over “ v unmatched”,

$$P_v = \prod_i P_{c_i} + \sum_i w_{vc_i} P_{c_i}^0 \prod_{j \neq i} P_{c_j} = \left(\prod_i P_{c_i} \right) \left(1 + \sum_i w_{vc_i} \frac{P_{c_i}^0}{P_{c_i}} \right) = \left(\prod_i P_{c_i} \right) \left(1 + \sum_i w_{vc_i} r_{c_i} \right).$$

Dividing, $r_v = P_v^0/P_v = (1 + \sum_i w_{vc_i} r_{c_i})^{-1}$. The division step $r_v = P_v^0/P_v = \prod_i P_{c_i} / [(\prod_i P_{c_i})(1 + S)] = 1/(1 + S)$ (with $S = \sum_i w_{vc_i} r_{c_i}$) is the algebraic identity machine-checked in `Matching.lean` (`cavity_step`). $\square$

Theorem 4. $\pi(T) = P_\rho$, and $\log \pi(T) = \sum_{v \in V} \log(1 + \sum_{c \text{ child of } v} w_{vc} r_c)$. Thus $\pi(T)$ is computed by the cavity recursion of Lemma 3, evaluated once per vertex from the leaves to the root.

Proof. $Z(T) = \sum_M \prod_e w_e = P_\rho$ by definition ($T_\rho = T$), and $Z(T) = \pi(T)$ by Corollary 2. Telescoping $P_v = (\prod_i P_{c_i})(1 + \sum_i w_{vc_i} r_{c_i})$ down the tree, every P_c appears once as a factor and once inside its parent's product, so $P_\rho = \prod_v (1 + \sum_c w_{vc} r_c)$ (leaves contribute the empty product = 1). Taking logs gives the stated sum. $\square$

4 Consequence and scope

Corollary 2 and Theorem 4 are the matching/permanent bridge *for the raw Laplacian weights* $w_{ij} = 1/(\deg i \deg j)$: the quantity this edge-weighted cavity recursion $r_v = (1 + \sum_c w_{vc} r_c)^{-1}$ telescopes — with the per-vertex log-sum — is *exactly* $\pi(T) = \text{per } L(T) / \prod \deg$ for every finite tree.

Relation to the R3 branch amplitudes (scope). The branch amplitudes $\log \Phi = \sum_v [\dots]$ used in R3 are *not* this raw cavity. They use the DEC-*dressed* recursion, in which each length-2 cherry is folded into the vertex weight $z(d, c) = 3/(3d + c)$ (dressed edge weight $z(d, c) z(d', c')$), together with the fixed ρ_B amplitude normalisation. The dressed per-node cavity rho0 is a matching cavity — but of the *dressed* tree, not of $L(T)$: numerically the naive raw-weight cavity of the explicit cherry tree does *not* equal rho0 (e.g. a single-cherry node gives $\text{rho0} = 1$ vs. raw $6/7$). That identification $\text{rho0} = Z_{\text{open}}/Z_{\text{tot}}(\text{realize})$ is the machine-checked companion result (`Bridge.lean/BridgeStep2.lean`, Steps 1–2). Tying $\log \Phi$ back to $\pi(T)$ for a real tree is *not* a fixed rescaling by a power of ρ_B (empirically $\log_{\rho_B}(\pi/\Phi)$ is not integer): it factors through the $p \rightarrow \infty$ cherry-hub limit (Step 4), which is open. So this note settles the raw-weight bridge $\pi(T) = Z(T)$; it does *not* by itself establish that the R3 dressed branch amplitude is the genuine Laplacian ratio.

What is machine-checked, and what remains. The algebraic recursion step of Lemma 3 (the passage $P_v, P_v^0 \Rightarrow r_v = 1/(1 + S)$) is proved in Lean (`cavity_step`, no `sorry`); both identities (H1) and (H2) are verified numerically to machine precision on all trees up to 10 vertices. A full Lean formalisation of (H1) — the permutation cycle-decomposition over an acyclic support, using Mathlib's `Matrix.permanent`, `SimpleGraph.IsAcyclic`, and `Equiv.Perm` cycle factors — and of the combinatorial bijection in Lemma 3 is a self-contained graph-theory development left as mechanical work; the proofs above are complete and standard. This is the honest boundary: the mathematics of the *raw-weight* matching/permanent bridge (H1, H2) is settled here, and its end-to-end proof-assistant encoding is the remaining formalisation *for that part*. The connection to the R3 dressed branch amplitude is a *separate* bridge: Steps 1–2 (cherry-folding, $\text{cav} = z \cdot \text{rho0}$ and $\text{rho0} = Z_{\text{open}}/Z_{\text{tot}}(\text{realize})$) are machine-checked; Step 3 (the dressed RTree as a weighted `SimpleGraph`) and Step 4 (the $\rho_B/p \rightarrow \infty$ hub-limit amplitude normalisation tying $\log \Phi \leq 0$ back to $\text{per } L(T) / \prod \deg$) remain open. $\Phi \leq 1$ (Conjecture 1) is not proved.